



(Don't Write
anything in this Area)

5.b) Mutations and its role in human evolution

Mutations are the sudden heritable change in the DNA of the human.

Types of Mutations are :-

Small scale Mutations :

(i) Single Nucleotide Mutation: change in only one base pair is observed.

(ii) Frame shift Mutation : $\rightarrow$ Addition or deletion of one or more bases impart the expression of entire gene.

(Eg) Sickle cell trait, it is a point Mutation; $SS = \text{normal}$; $ss = \text{sickle cell}$.

Large scale Mutation

i) Deletion of the set of genes from the chromosome.

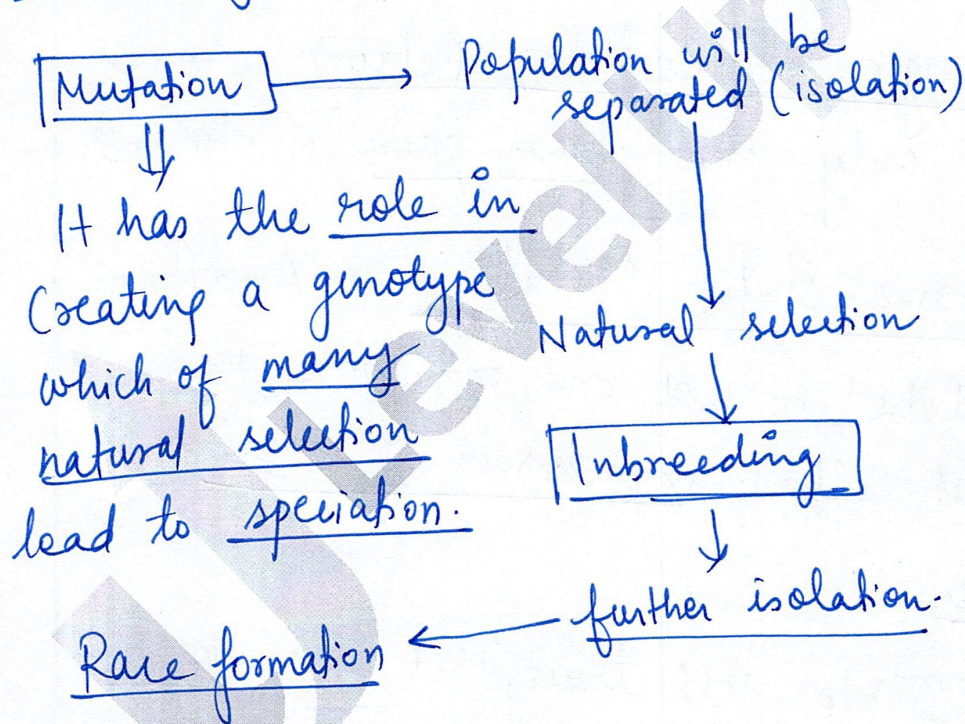
(ii) Addition of the genes.

(iii) Inversion of the genes.

(iv) Translocation of genes.

(Eg) → Various genetic syndromes.

Role of Mutation in Human evolution



Hence, the mutation is the one which has the evolutionary significance in the human history.